Water

Life on earth began in the seas and water is essential for the survival of all forms of life. About three-fourth of the earth's surface is covered with water (hydrosphere). Water is found in three states: gaseous, liquid, and solid.

There are two types of water on Earth. They are the Fresh water (rivers, lakes, ponds) and the Salt water (seas and oceans). Based on the dissolved salts, water can be hard water (sulphates/nitrates of Calcium/Magnesium) or soft water. If hardness can be removed by boiling, it is temporary hard water, and if boiling does not help, it is permanent hard water.

Essential properties of water

- Water is one of the main agents in Pedogenesis (soil formation).
- It is the medium for several different ecosystems.
- It is present as moisture in the atmosphere and the outer layers of the lithosphere and is uneven in distribution on the earth.
- Water is heavier than air and imparts greater buoyancy to the aquatic medium. This enables organism to float at variable levels.
- Water has high heat capacity and latent heat, due to which it can withhold large amounts of heat. Thus, oceans and lakes tend to maintain a relatively constant temperature, and the biosphere is relatively thermostable.
- Water is physically unique because it is less dense as a solid (ice) than as a liquid.
- When water freezes (0°C), it contracts. The maximum density of liquid water occurs at 4°C. Below that, it expands markedly. This enables ice to float on the top of water bodies. Hence, only the surface of water bodies will freeze, while below the surface, water will be in liquid form, sustaining life (**Fig. 10.3).**
- Water is considered as the Universal solvent. It is the main medium by which chemical constituents are transported from abiotic components to the living components of an ecosystem.
- Water has high surface tension. This allows pollen, dust, and even water striders to remain at the surface of a water body even though they are denser than the water.

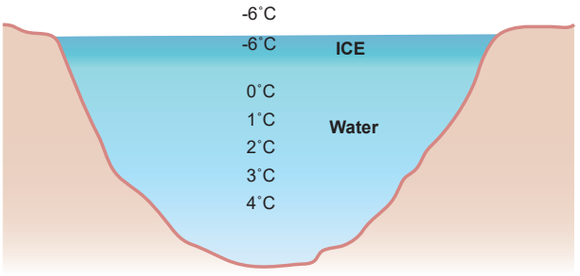


Fig. 10.3 Thermal zones in cold water bodies

Soil

It is a mixture of organic matter, minerals, gases, liquids and organisms that together support life. The soil zone is known as **Pedosphere**. Soil is formed from rocks which are the parent materials of soil, by weathering and is called embryonic soil (Pedogenesis).

It has four major functions-

- medium for plant growth
- means for water storage and purification
- modifier of earth's atmosphere
- habitat for many organisms, which in turn modify the soil

Soil is formed of many horizontal layers called as Soil Profile.

Properties of Soil

1. **Texture of soil** – The texture of soil is determined by the size of the soil particles. The types of soil include sand, silt and clay on the basis of their size differences.
2. **Porosity** – The space present between soil particles in a given volume of soil are called pore spaces. The percentage of soil volume occupied by pore space or by the interstitial spaces is called porosity of the soil.
3. **Permeability of soil**-The characteristic of soil that determines the movement of water through pore spaces is known as soil permeability. Soil permeability is directly dependent on the pore size. Water holding capacity of the soil is inversely dependent on soil porosity.

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